CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (CL) Examination

May 2013

CL421 Advanced Structural Analysis

Date: 11.05.2013, Saturday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Enlist and explain various forces acting on the bridge. **[07]**

Q - 2 Calculate design bending moment and design shear force for following data of deck slab. **[14]**

Carriage way – Two lane (7.5 m wide)

Clear span – 6 m

Width of footpath – 1 m on either side

Wearing coat – 80 mm

Width of bearing = 400 mm

Loading – IRC Class AA tracked vehicle

Material – Concrete grade M 25 & Fe 415 grade HYSD bars

OR

Q - 2 Calculate design bending moment and design shear force for following data of deck slab. **[14]**

Clear span – 4 m

Width of footpath – 600 mm on either side

Wearing coat – 80 mm

Loading – IRC class A

Material – Concrete grade M 25 & Fe 415 grade HYSD bars

Q - 3 (a) Explain analysis of slabs spanning in two directions. **[07]**

(b) Explain any one method from the following for the analysis of a typical tee beam and slab deck bridge. **[07]**

1) Morrice Little Method
2) Hendry Jaegar Method
3) Corbon's Method

SECTION – II

Q - 4 Enlist method of non – linear analysis and explain any one in detail. [07]

Q - 5 (a) **Figure 1** shows an assembly of three springs. Derive force displacement relationship for member AB and tangent stiffness matrix for member AB. [08]

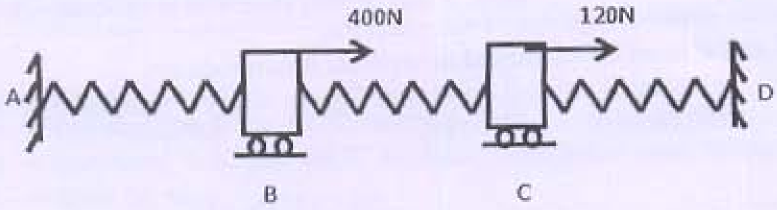

Figure: 1

For AB spring nonlinear F-D relation is $F = 100e - 10e^2$ in which e is elongation in mm and F is force in Newton.

(b) Derive an expression for Meridional thrust and hoop force for a spherical dome subjected to concentrated load at crown. [06]

OR

(b) Derive an expression for Meridional thrust and hoop force for a conical dome subjected to uniformly distributed load at crown. [06]

Q - 6 (a) Enlist the various types of dome. [04]

List the loads acting on dome.

(b) Design a conical roof for a hall having diameter of 18 m and the rise of the dome is 4 m. Take the live load as 1600 N/m^2. Also give the design of ring beam. Use M 20 grade concrete and Fe 415 steel. [10]

OR

(b) Design a spherical dome for a temple hall of 8 m diameter. The rise of the dome is to be 1.6 m. The live load may be taken as 750 N/m^2 and floor finishes as 250 N/m^2. The dome carries a lantern load of 10 kN concentrated at the crown. Design the ring beam also. Use M 20 grade concrete and Fe 415 steel. [10]

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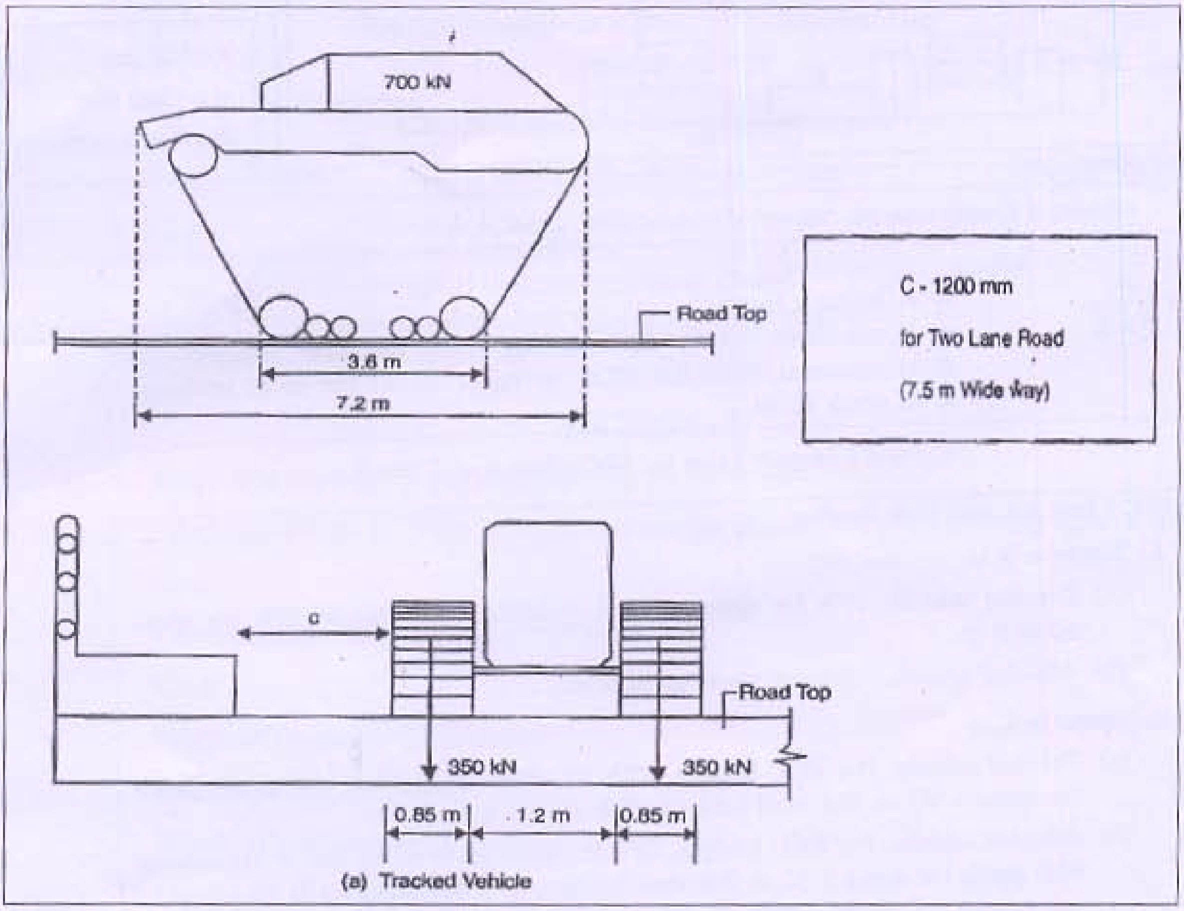


Figure: 1 Class AA (Tracked)

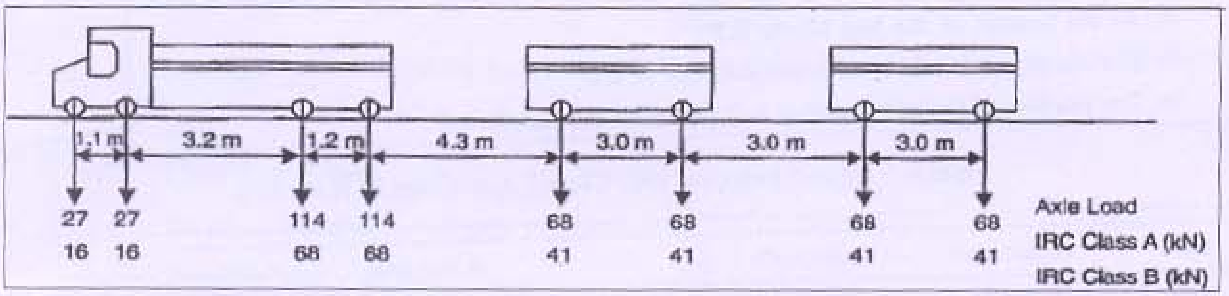


Figure: 2 Side View Details (IRC Class A & B)

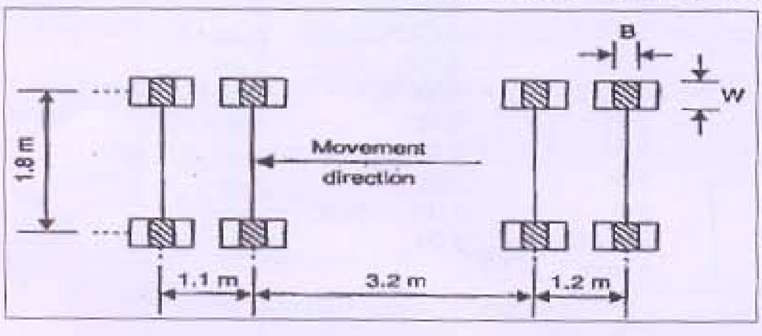
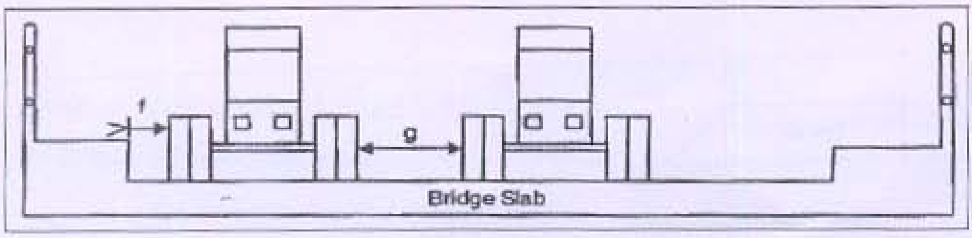


Figure: 3 Wheel impressions on road (IRC Class A & B)

Axle load	Ground contact area	
	B (mm)	W (mm)
114	250	500
68	200	380
41	150	300
27	150	200
16	125	175

Table: 1



f = 150 mm
g = 1200 mm
(for 7.5 m wide way)

Figure: 4 Transverse placement of wheels (IRC Class A & B)

Figure: 5

$$I_f = \frac{A}{B+L}$$

where

I_f = impact factor
A = constant, 4.5 for RCC bridges, 9.0 for steel bridges
B = constant, 6.00 for RCC bridges, 13.50 for steel bridges
L = span in m.

Figure: 6 Impact value for IRC Class A or Class B

For IRC Class AA and 70R loading

1. **Spans < 9 m**
   (a) *Tracked vehicle.* 25% for spans up to 5 m linearly reducing to 10% for spans up to 9 m.
   (b) *Wheeled vehicle.* 25% for spans up to 9 m.
2. **Spans > 9 m**
   (a) *Tracked vehicle.* For RCC bridges, 10% for spans up to 40 m and as per graph for spans > 40 m. For steel bridges, 10% for all spans.
   (b) *Wheeled vehicle.* For RCC bridges, 25% for spans up to 12 m, and in accordance with graph for spans > 12 m. For steel bridges, 25% for spans up to 23 m and as per graph (IRC 6) for spans exceeding 23 m.

Appropriate impact factors as mentioned below need be considered for substructures as well.

- At the bottom of the bed block: 0.5
- For the top 3 m of the substructure: 0.5 to 0.0
- For portion of the substructure > 3 m below the block: 0.0

Figure: 7 Impact value for IRC Class AA or Class 70 R loading

B/l	α for sss*	α for cs*	B/l	α for sss	α for cs
0.1	0.40	0.40	1.1	2.60	2.28
0.2	0.80	0.80	1.2	2.64	2.36
0.3	1.16	1.16	1.3	2.72	2.40
0.4	1.48	1.44	1.4	2.80	2.48
0.5	1.72	1.68	1.5	2.84	2.48
0.6	1.96	1.84	1.6	2.88	2.52
0.7	2.12	1.96	1.7	2.92	2.56
0.8	2.24	2.08	1.8	2.96	2.60
0.9	2.36	2.16	1.9	3.00	2.60
1.0	2.48	2.24	2.0 and above	3.00	2.60

* sss = simply supported slab, cs = continuous slab

Table: 2 Values of α for slabs (IRC 21)